SynthMicro — Whole Blood Smear XAI Report

Analysis ID: 6af71b5caef54bbda46634a23d664347
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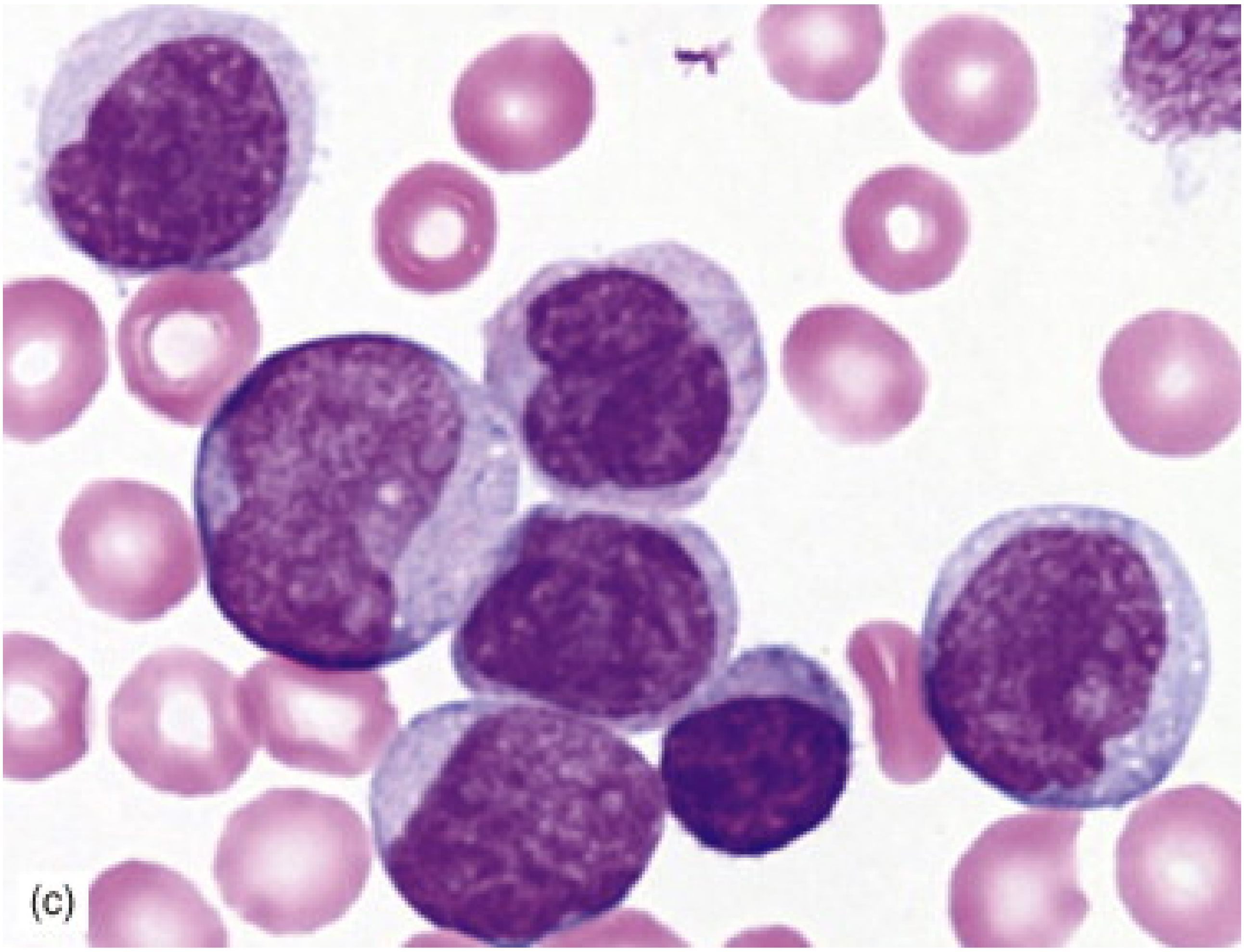
Pipeline: Cellpose segmentation → 384×384 cell extraction → ResNet50 classification → Grad-CAM → SHAP → explanatory report.

Measurement	Result
Cellpose objects	23
Nucleated-cell candidates	23
Myeloblast-like candidates	3
Myeloblast-like proportion	13.04%

Screening interpretation: the proportion above is a research-oriented model indicator based on candidate cells selected by the notebook's purple-score rule. It is not a cancer probability and does not establish AML or another diagnosis.

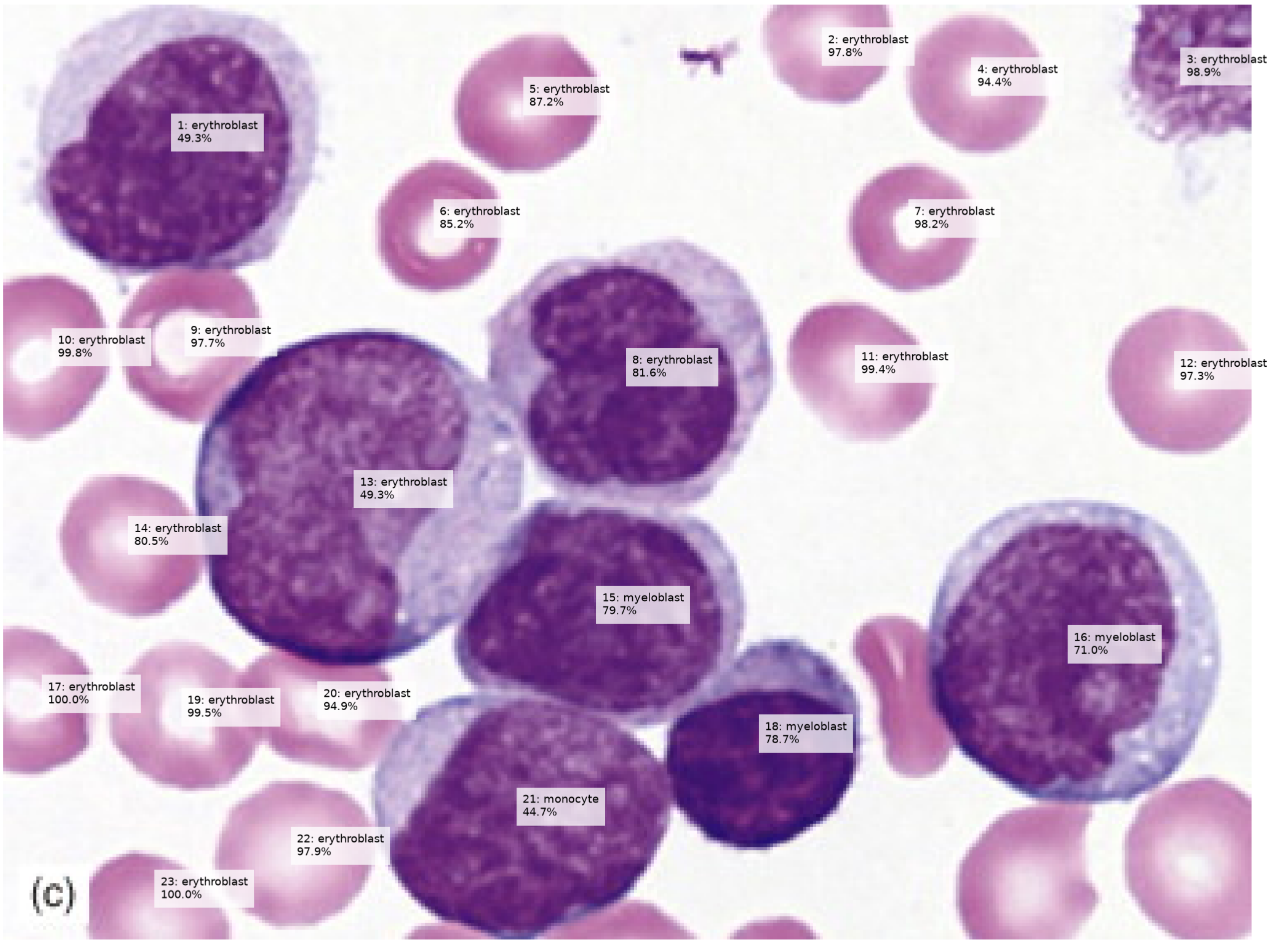
Whole Smear



Original uploaded blood-smear image analyzed by Cellpose.

Segmentation and Candidate Overlay

Whole Smear — Candidate Cell Classification



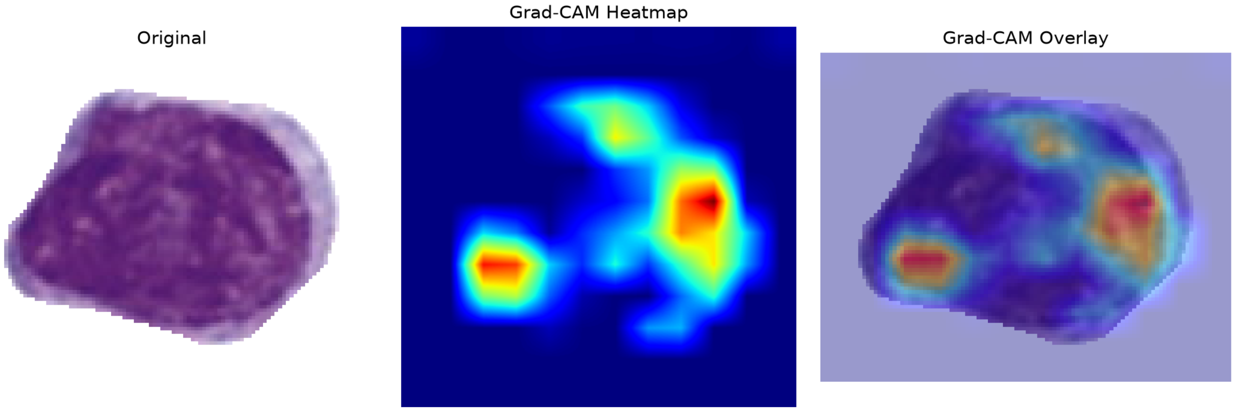
Cell-Level Results

ID	Area	Purple	Class	Confidence
1	5883	54.0	erythroblast	49.3%
2	935	39.7	erythroblast	97.8%
3	1361	51.5	erythroblast	98.9%
4	1389	43.8	erythroblast	94.4%
5	1436	54.0	erythroblast	87.2%
6	1187	52.6	erythroblast	85.2%
7	1142	51.1	erythroblast	98.2%
8	5533	54.9	erythroblast	81.6%
9	1566	47.7	erythroblast	97.7%
10	1281	48.2	erythroblast	99.8%
11	1357	43.3	erythroblast	99.4%
12	1487	43.1	erythroblast	97.3%
13	7935	54.4	erythroblast	49.3%
14	1453	45.0	erythroblast	80.5%
15	4541	62.1	myeloblast	79.7%
16	6677	53.0	myeloblast	71.0%
17	950	47.1	erythroblast	100.0%
18	2940	60.2	myeloblast	78.7%
19	1554	40.8	erythroblast	99.5%
20	1315	44.5	erythroblast	94.9%
21	5261	60.6	monocyte	44.7%
22	1650	37.1	erythroblast	97.9%
23	783	42.6	erythroblast	100.0%

Grad-CAM Explanations

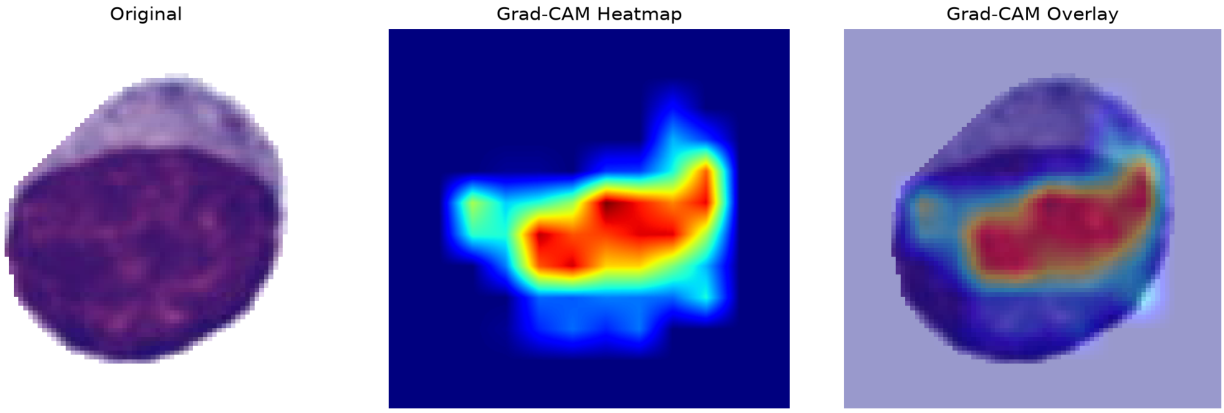
Grad-CAM shows regions contributing to the selected model class. It is an explanation of model behavior, not a direct measurement of a biological structure.

Cell 15 — myeloblast (79.7%)



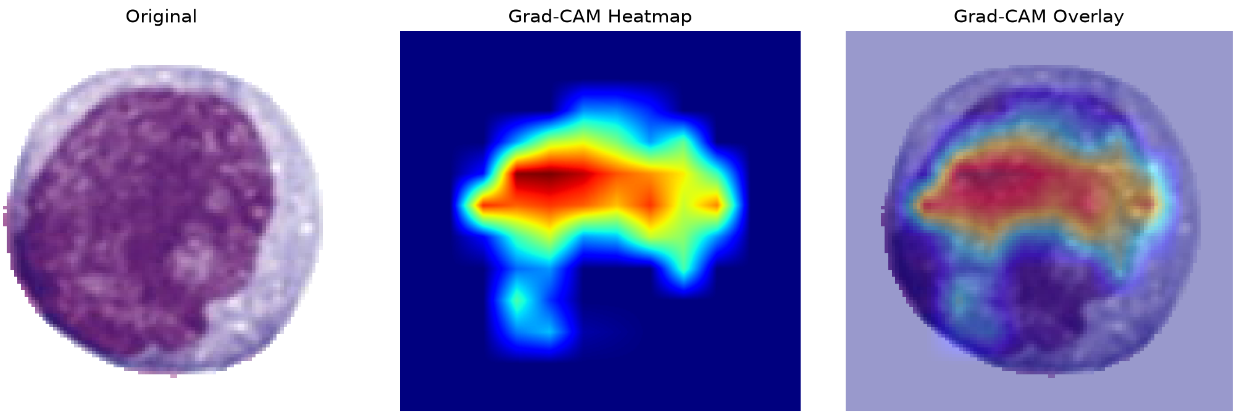
The ResNet50 model classified this segmented candidate as a myeloblast with 79.7% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 18 — myeloblast (78.7%)



The ResNet50 model classified this segmented candidate as a myeloblast with 78.7% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 16 — myeloblast (71.0%)



The ResNet50 model classified this segmented candidate as a myeloblast with 71.0% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

SHAP Explanations

SHAP attribution magnitude summarizes the magnitude of pixel-level contribution toward the selected class. It is a model explanation, not a clinical measurement.

SHAP output was unavailable for this analysis.

Interpretation and Limitations

• The classifier has five closed-set classes: basophil, erythroblast, monocyte, myeloblast and segmented neutrophil.

• RBCs and unknown cell types are not explicit classes and can therefore be forced into one of the five outputs.

• Cellpose segmentation quality affects every downstream result.

• Stain, microscope, illumination and acquisition differences can cause domain shift.

• Confidence is not a calibrated probability of disease.

• A myeloblast-like prediction is not equivalent to an AML diagnosis.

• Clinical assessment requires appropriate hematopathology and laboratory testing.